Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Payment Confirmation System

A PROJECT REPORT

for

Agile Development Process and Devops (ISWE406L) – C2+TC2

Winter Semester 2025-2026

In

Integrated M.Tech - Software Engineering

By

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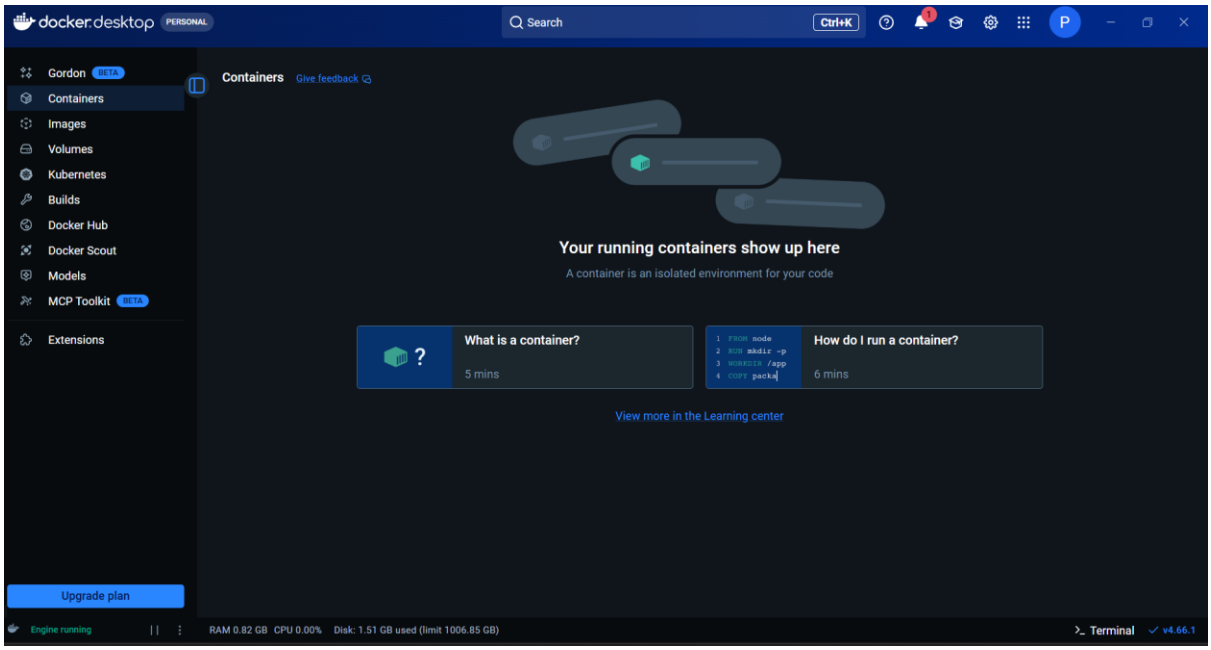
Under the Guidance of

Dr. Bala Ganesh N

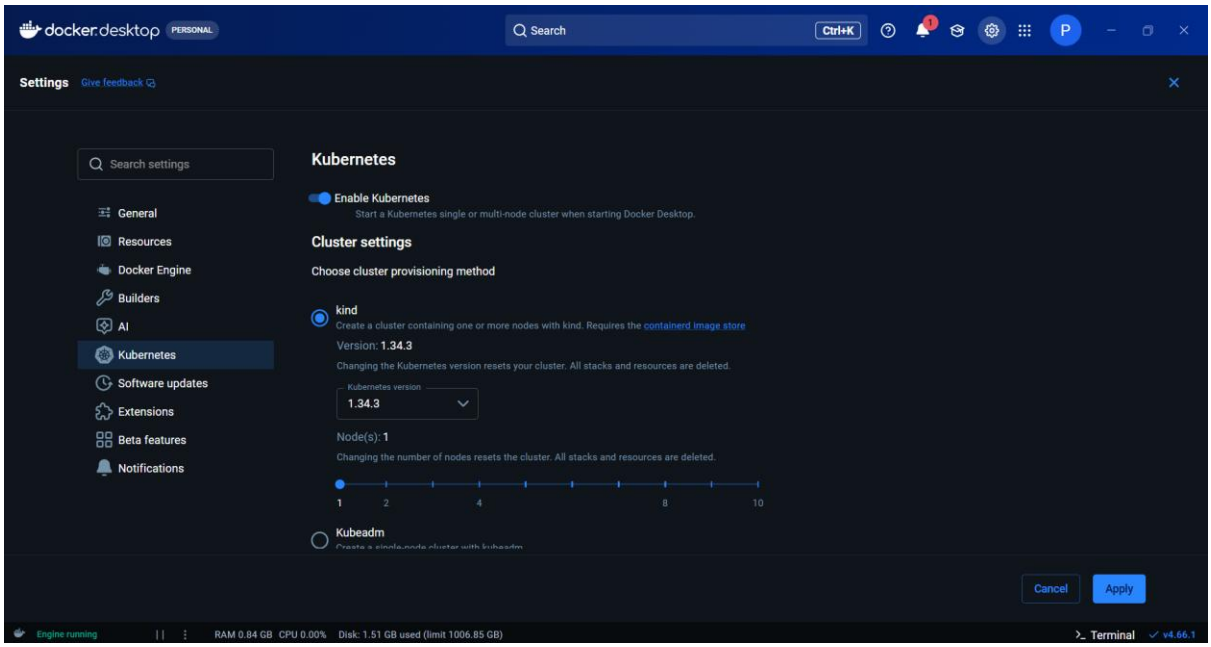
Assistant Professor

Step 1:

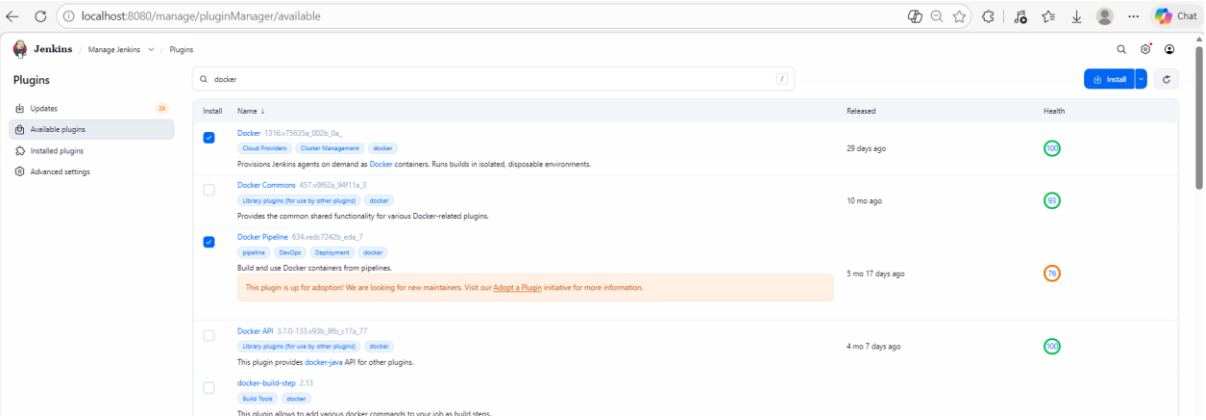
Install Docket Desktop

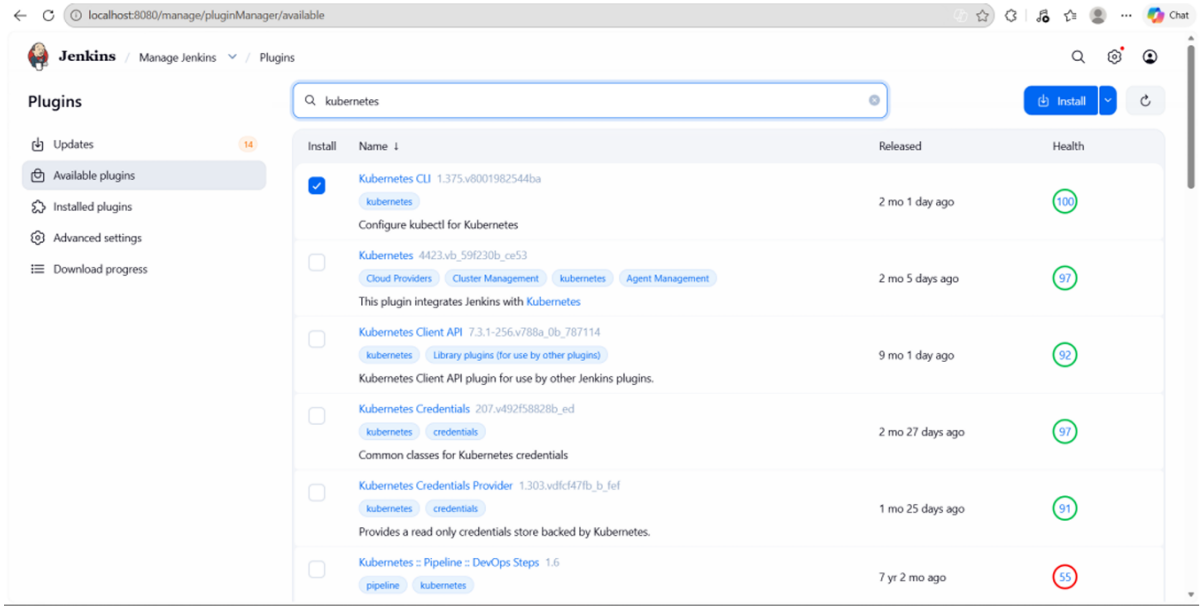


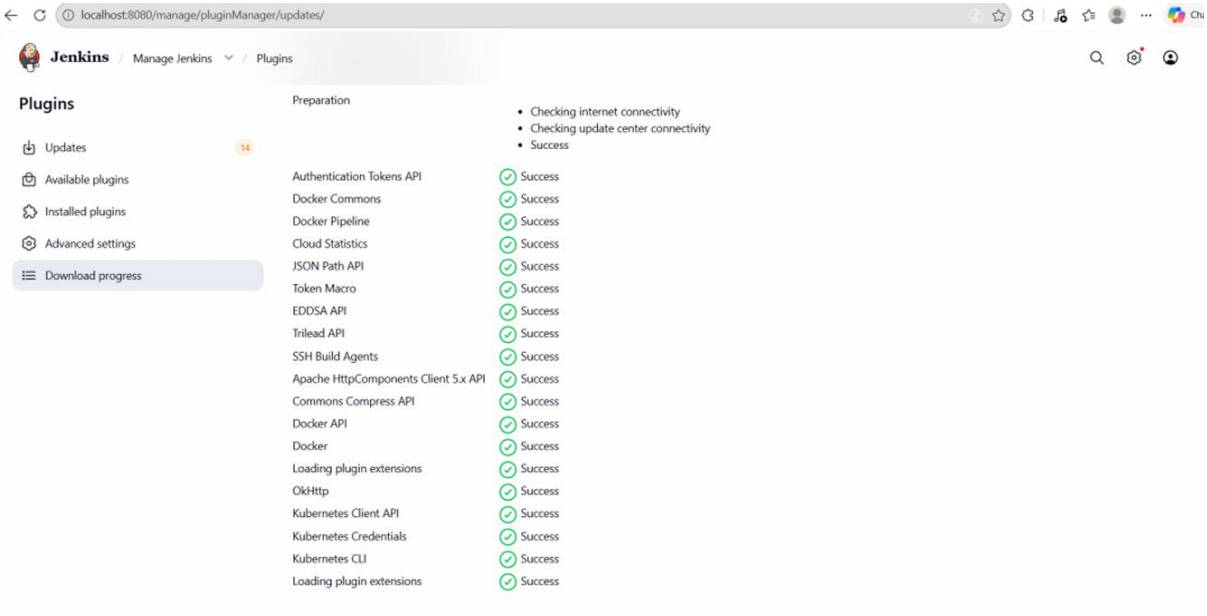
Enable Kubernets



Enable three plugins (Docker, Docker Pipeline, Kubernetes CLI) and install them:


localhost:8080/manage/pluginManager/available
Jenkins / Manage Jenkins / Plugins
Plugins
docker
Install
Updates
Available plugins
Installed plugins
Advanced settings
Install
Name
Released
Health
Docker 1316.v75635a_002b_0a_
Cloud Providers Cluster Management docker
Provisions Jenkins agents on demand as Docker containers. Runs builds in isolated, disposable environments.
29 days ago
100
Docker Commons 457.v0f62a_94f11a_3
Library plugins (for use by other plugins) docker
Provides the common shared functionality for various Docker-related plugins.
10 mo ago
93
Docker Pipeline 634.vedc7242b_eda_7
pipeline DevOps Deployment docker
Build and use Docker containers from pipelines.
This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.
5 mo 17 days ago
76
Docker API 3.7.0-133.v93b_8ffb_c17a_77
Library plugins (for use by other plugins) docker
This plugin provides docker-java API for other plugins.
4 mo 7 days ago
100
docker-build-step 2.13
Build Tools docker


localhost:8080/manage/pluginManager/available
Jenkins / Manage Jenkins / Plugins
Plugins
kubernetes
Install
Updates 14
Available plugins
Installed plugins
Advanced settings
Download progress
Install
Name
Released
Health
Kubernetes CLI 1.375.v8001982544ba
kubernetes
Configure kubectl for Kubernetes
2 mo 1 day ago
100
Kubernetes 4423.vb_59f230b_ce53
Cloud Providers Cluster Management kubernetes Agent Management
This plugin integrates Jenkins with Kubernetes
2 mo 5 days ago
97
Kubernetes Client API 7.3.1-256.v788a_0b_787114
kubernetes Library plugins (for use by other plugins)
Kubernetes Client API plugin for use by other Jenkins plugins.
9 mo 1 day ago
92
Kubernetes Credentials 207.v492f58828b_ed
kubernetes credentials
Common classes for Kubernetes credentials
2 mo 27 days ago
97
Kubernetes Credentials Provider 1.303.vdfcf47fb_b_fef
kubernetes credentials
Provides a read only credentials store backed by Kubernetes.
1 mo 25 days ago
91
Kubernetes :: Pipeline :: DevOps Steps 1.6
pipeline kubernetes
7 yr 2 mo ago
55


localhost:8080/manage/pluginManager/updates/
Jenkins / Manage Jenkins / Plugins
Plugins
Updates 14
Available plugins
Installed plugins
Advanced settings
Download progress
Preparation
Checking internet connectivity
Checking update center connectivity
Success
Authentication Tokens API Success
Docker Commons Success
Docker Pipeline Success
Cloud Statistics Success
JSON Path API Success
Token Macro Success
EDDSA API Success
Trilead API Success
SSH Build Agents Success
Apache HttpComponents Client 5.x API Success
Commons Compress API Success
Docker API Success
Docker Success
Loading plugin extensions Success
OkHttp Success
Kubernetes Client API Success
Kubernetes Credentials Success
Kubernetes CLI Success
Loading plugin extensions Success

Step 2 - Create Simple Java Application (in local machine)

Create App.java

```java
public class App {

    // Method to validate payment
    public static String confirmPayment(double amount, String status) {

        if (amount <= 0) {
            return "Invalid Payment Amount";
        }

        if (status.equalsIgnoreCase(anotherString: "SUCCESS")) {
            return "Payment Confirmed";
        } else {
            return "Payment Failed";
        }
    }

    Run | Debug
    public static void main(String[] args) {
        System.out.println(confirmPayment(amount: 500, status: "SUCCESS"));
    }
}
```

Create AppTest.java

```java
import static org.junit.jupiter.api.Assertions.assertEquals;
import org.junit.jupiter.api.Test;

public class AppTest {

    // Test for successful payment
    @Test
    void testPaymentSuccess() {
        assertEquals("Payment Confirmed", App.confirmPayment(500, "SUCCESS"));
    }

    // Test for failed payment
    @Test
    void testPaymentFailure() {
        assertEquals("Payment Failed", App.confirmPayment(500, "FAILED"));
    }

    // Test for invalid amount
    @Test
    void testInvalidAmount() {
        assertEquals("Invalid Payment Amount", App.confirmPayment(0, "SUCCESS"));
    }
}
```

Create pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
 <dependencies>
  </dependency>
 </dependencies>

 <build>
  <plugins>

   <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-compiler-plugin</artifactId>
    <version>3.11.0</version>
   </plugin>

   <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-surefire-plugin</artifactId>
    <version>3.2.5</version>
    <configuration>
     <useModulePath>false</useModulePath>
    </configuration>
   </plugin>

   <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-jar-plugin</artifactId>
    <version>3.4.1</version>
    <configuration>
     <archive>
      <manifest>
       <mainClass>App</mainClass>
```

Step 3 - Build Locally using Maven (Test)

Inside project folder:

mvn clean test:

```
PROBLEMS 11   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\Pradheep\Desktop\payment confirmation> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\payment confirmation\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\payment confirmation\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] No tests to run.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.148 s
[INFO] Finished at: 2026-04-11T14:30:01+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Pradheep\Desktop\payment confirmation> mvn clean package
```

mvn clean package

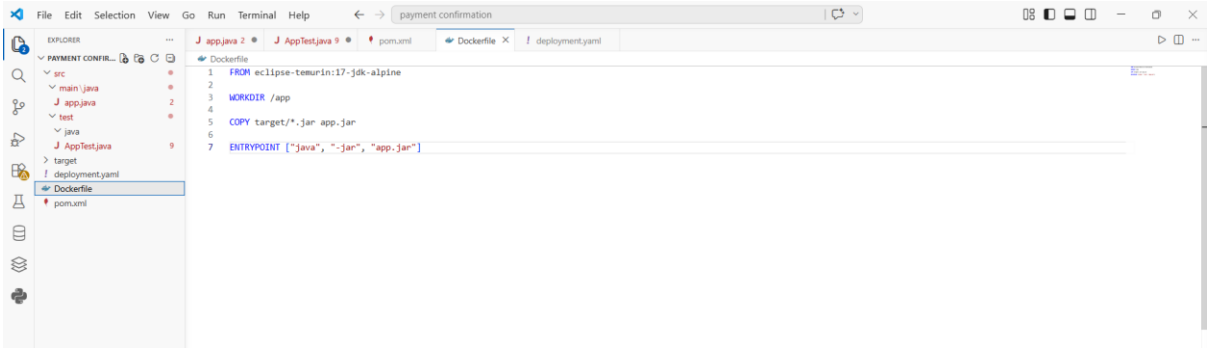
```
PS C:\Users\Pradheep\Desktop\payment confirmation> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.demo:test-java-app >-----------------------
[INFO] Building test-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ test-java-app ---
[INFO] Deleting C:\Users\Pradheep\Desktop\payment confirmation\target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\payment confirmation\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ test-java-app ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ test-java-app ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Pradheep\Desktop\payment confirmation\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ test-java-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO] Nothing to compile - all classes are up to date
[INFO] --- surefire:3.2.5:test (default-test) @ test-java-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ test-java-app ---
[INFO] Building jar: C:\Users\Pradheep\Desktop\payment confirmation\target\test-java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.287 s
[INFO] Finished at: 2026-04-11T14:30:13+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Pradheep\Desktop\payment confirmation>
```

It generates: target/ test-java-app-1.0.jar

Name	Date modified	Type	Size
src	11-04-2026 14:23	File folder	
target	11-04-2026 14:30	File folder	
deployment.yaml	11-04-2026 14:27	Yaml Source File	1 KB
Dockerfile	11-04-2026 14:28	File	1 KB
pom.xml	11-04-2026 14:27	Microsoft Edge HT...	2 KB

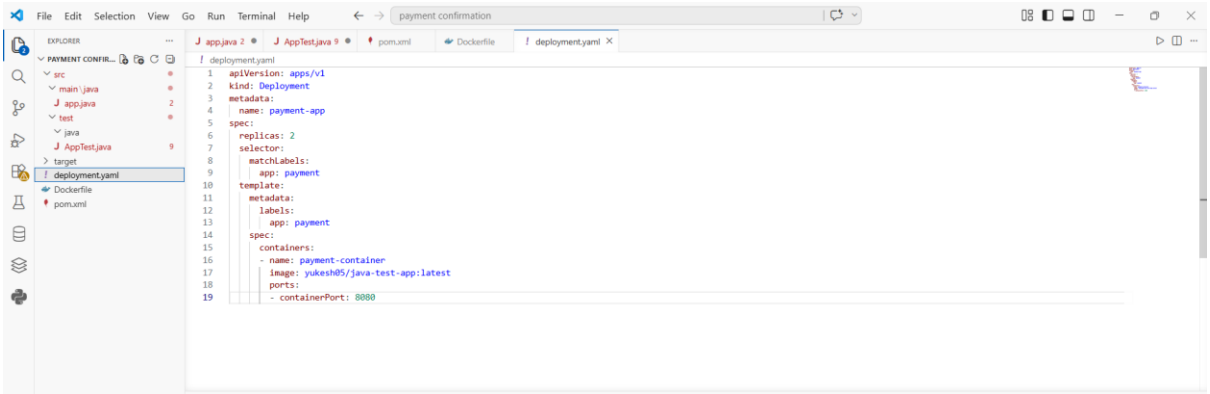
Name	Date modified	Type	Size
classes	11-04-2026 14:30	File folder	
generated-sources	11-04-2026 14:30	File folder	
generated-test-sources	11-04-2026 14:30	File folder	
maven-archiver	11-04-2026 14:30	File folder	
maven-status	11-04-2026 14:30	File folder	
test-java-app-1.0.jar	11-04-2026 14:30	Executable Jar File	2 KB

Step 4 - Create Dockerfile



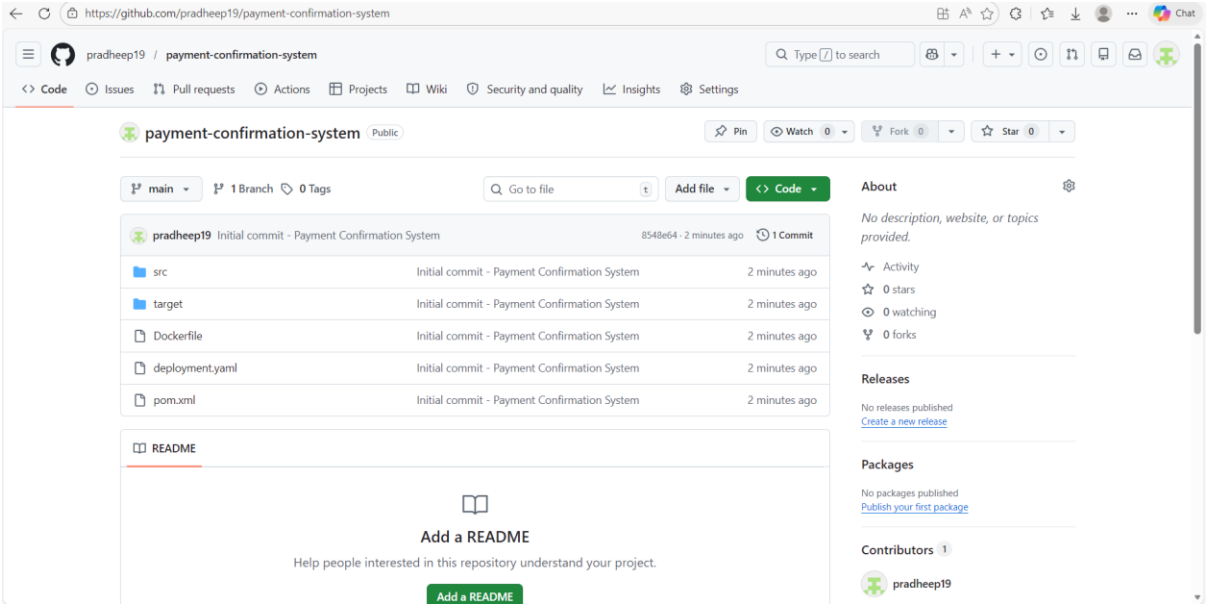
Step 5 - Create Kubernetes Deployment File

deployment.yaml

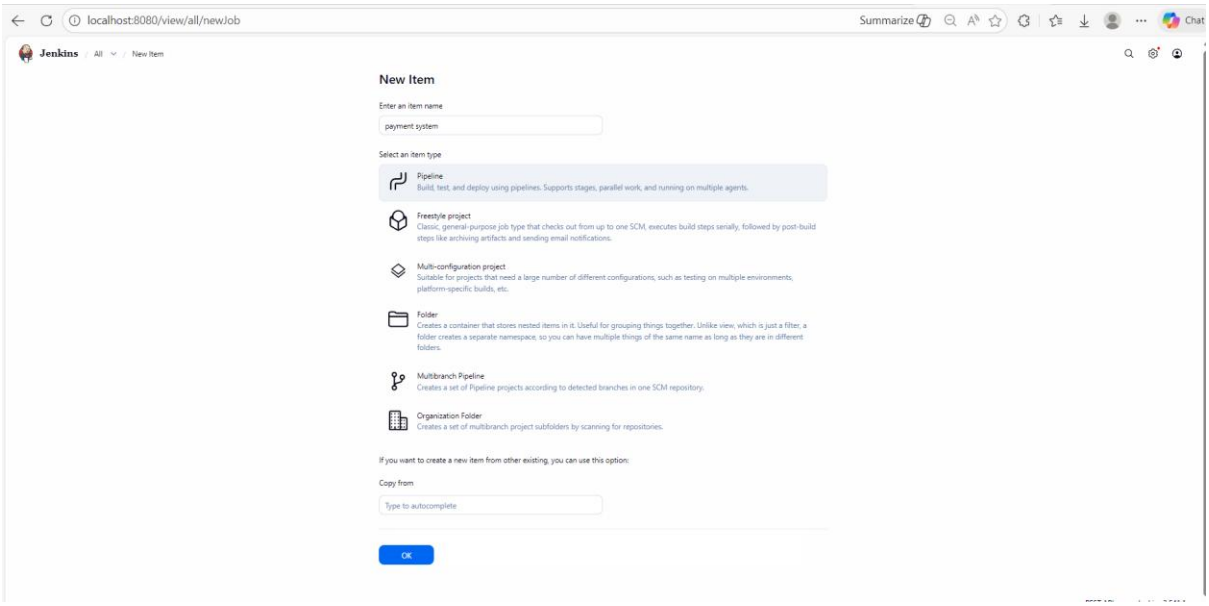


Step 6 - Push Code to GitHub

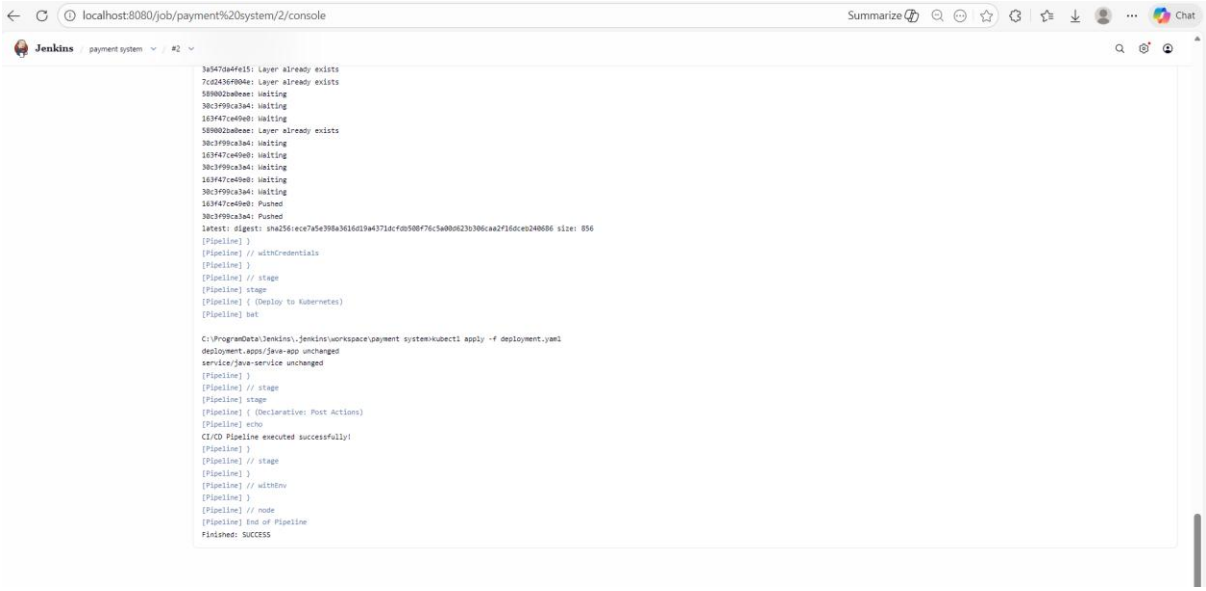
```
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/test-java-app-1.0.jar
PS C:\Users\Pradheep\Desktop\payment confirmation> git remote add origin https://github.com/pradheep19/payment-confirmation-system
PS C:\Users\Pradheep\Desktop\payment confirmation> git branch -M main
PS C:\Users\Pradheep\Desktop\payment confirmation> git push -u origin main
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 16 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (21/21), 2.98 KiB | 763.00 KiB/s, done.
Total 21 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/pradheep19/payment-confirmation-system
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Pradheep\Desktop\payment confirmation>
```

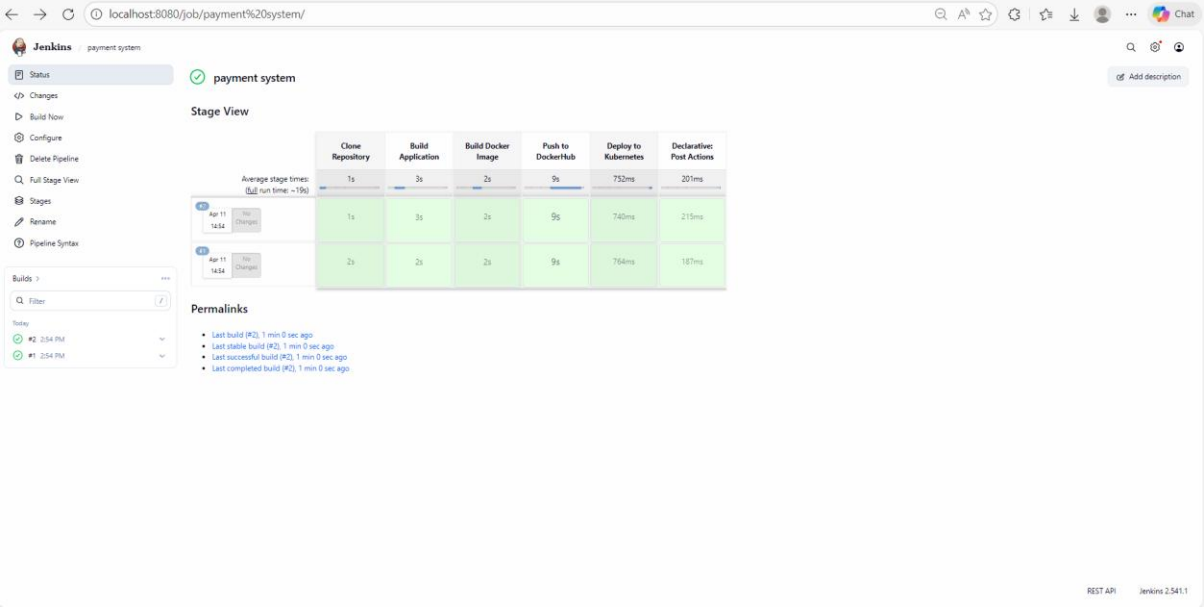


Step 7 - Create Jenkins Pipeline Job



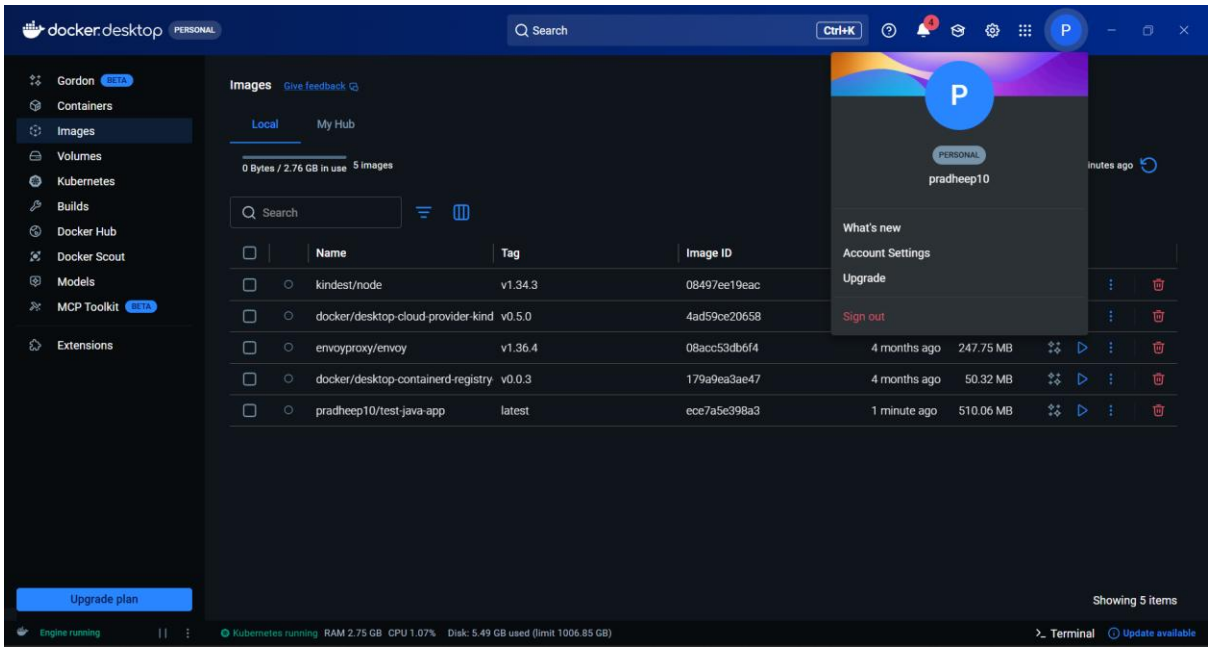
Console output:



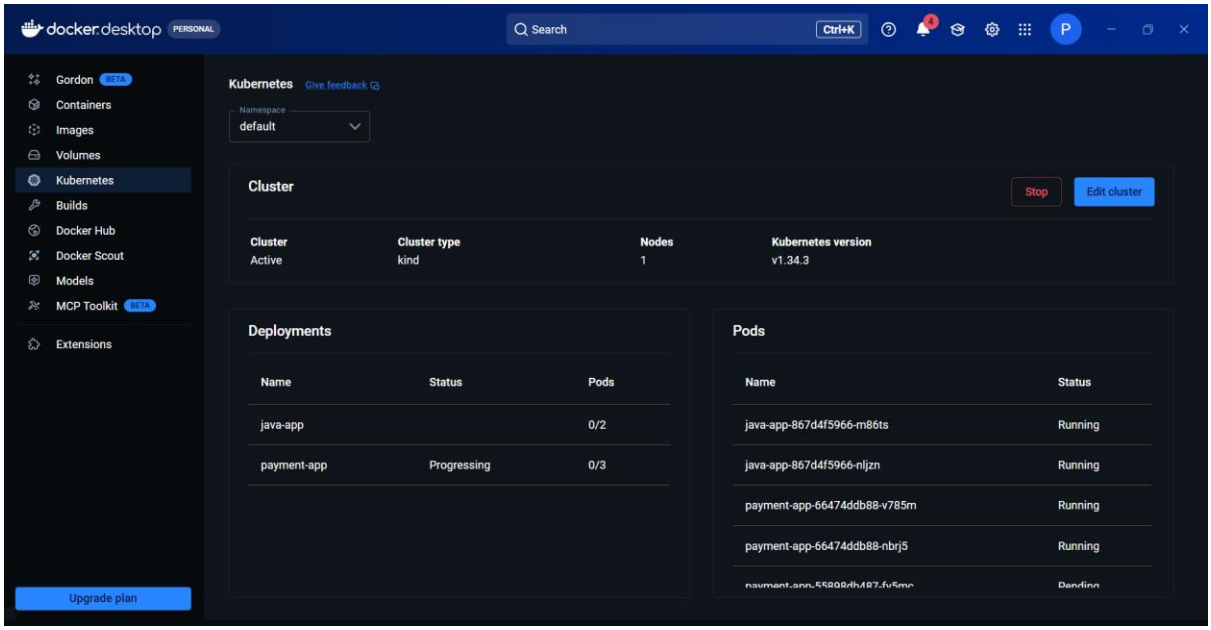


Step 12 - Verify Docker and Kubernetes Deployment

Docker Image



Kubernetes




```
C:\Users\Pradheep\Desktop\payment confirmation>kubectl get svc
NAME           TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
java-service   NodePort    10.96.120.178   <none>        80:30008/TCP   9s
kubernetes     ClusterIP   10.96.0.1       <none>        443/TCP        5d4h

C:\Users\Pradheep\Desktop\payment confirmation>
```